CAPSTONE PROJECT REVIEW PROGRESS RECORD

Review Stage:	[ ] Review–1 [ ] Review–2 **[X] Review–3 (Final)**	**Date of Review:**	**6 November 2026**
Project Title:	Sensitivity-Guided Layer-Wise Approximate Multiplier Mapping for FPGA-Based CNN Accelerators	**Cohort:**	2024–2028 (Batch B4)
Project Guide:	Dr. Arun Nimmala, Associate Professor, Dept. of ECE	**Target HW:**	PYNQ-Z2 (Xilinx Zynq-7020)

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Brief Description of the Project

Briefly explain the premise of the project, the problem being addressed and the proposed approach. (less than 200 words)

Convolutional Neural Network (CNN) edge accelerators are constrained by the high area and power dissipation of Multiply-Accumulate (MAC) datapaths, where 8x8 exact multipliers dominate silicon cost (68 LUTs on Xilinx Zynq-7020). While approximate computing trades controlled arithmetic error for hardware savings, uniform approximation causes severe accuracy degradation or negligible gains. Existing layer-wise mapping methods rely on computationally prohibitive evolutionary search (ALWANN: 7.5 CPU-days) or GPU retraining loops (TRAM). This project proposes a lightweight, single-pass sensitivity metric computed in 0.2 seconds combining static weight variance and dynamic loss gradients via a 256-sample calibration probe without retraining. Layers are ranked such that error-vulnerable early layers receive exact multipliers while noise-resilient deep layers receive aggressive approximate multipliers. The heterogeneous design is mapped onto synthesizable Verilog Processing Elements (PEs) and deployed on a PYNQ-Z2 FPGA board via AXI4-Stream DMA, achieving up to 32.5% multiplier area reduction while preserving baseline classification accuracy.

Progress Achieved up to This Review

Briefly mention the work completed, implementation/simulation/experimental progress, results obtained and other major milestones achieved up to this review.

- **NSGA-II Multi-Objective Optimization Benchmark:** Implemented evolutionary search across all 256 configurations optimizing Accuracy Loss vs. LUT Cost. Confirmed our sensitivity-guided assignment lands at the Pareto knee in 0.2s, achieving a >3,000,000x speedup over genetic search (ALWANN: 7.5 CPU-days).
- **Complete Synthesizable Accelerator RTL:** Designed full Verilog datapath (axm_8x8.v, mac_unit.v, conv_pe.v, layer_controller.v, cnn_accelerator_top.v) with 2-bit runtime mode control and AXI4-Stream interfaces. Verified across 262,144 exhaustive test vectors with 100% pass (0 mismatches).
- **Automated Vivado Block Design:** Developed vivado_block_design.tcl automating Zynq-7020 PS/PL integration, AXI SmartConnect, and AXI DMA engine (MM2S & S2MM channels) clocked at 100 MHz.
- **PYNQ-Z2 On-Board Deployment Pipeline:** Extracted INT8 quantized weights (weights_int8.npy), built Python on-board driver (pynq_inference.py), and authored an interactive 7-cell Jupyter notebook (pynq_cnn_demo.ipynb) for hardware telemetry.
- **Publication Readiness & Code Release:** Generated all 7 IEEE publication figures, drafted 6-page IEEE manuscript, compiled comprehensive 20-slide presentation deck with embedded speaker notes, and published repository to GitHub.

Signature of Team Leader: _____

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